Design and Implementation of ECTS in Sri Lanka

[Draft 2]

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Design and Implementation of an Electronic Cargo Tracking System in Sri Lanka

1. Project Background

1.1. Project Overview

Government of Sri Lanka is implementing the SASEC Port Access Elevated Highway Project, to provide a direct link to the Colombo Port from the Colombo-Katunayake Expressway via the New Kelani Bridge. This project is aimed to (i) reduce traffic congestion in the densely populated areas of Colombo, (ii) improve connectivity to Colombo Port and (iii) facilitate regional trade.

In support of the SASEC Port Access Elevated Highway Project, ADB is also implementing technical assistance "Supporting Trade Logistics Facilitation" to help Sri Lanka make progress toward meeting its commitments under the WTO Trade Facilitation Agreement. Lack of modern risk management practices and physical examination of a high share of cargo at off-site customs examination facilities are among the reasons that slow down clearance of cargo and contribute to traffic congestion within city limits. Electronic cargo tracking system (ECTS) for the monitoring and tracking of cargo movement between the Port and various examination centres and bonded warehouses will help mitigate some of the issues by effectively monitoring such movements and concurrently facilitating legitimate trade.

1.2. Project Objectives

The objective of the project is to support the Sri Lanka Customs Department in designing and implementing an electronic cargo tracking system (ECTS) for monitoring and tracking the movement of cargo between Colombo Port and various examination centres and bonded warehouses. This includes:

(1) Overall technical design of ECTS, including the hardware and software components of the system,
(2) Business processes to be adopted for the operation of ECTS, and
(3) How it will integrate with existing and future customs clearance system

1.3. Approach and Methodology

This approach and methodology followed for the design and implementation of ECTS in Sri Lanka consisted of a 4-part process:

I. Assessment of regulatory and economic environment in Sri Lanka

The first step in conducting a feasibility study was to assess the regulatory and economic conditions in Sri Lanka in the context of movement of cargo between Colombo Port and the examination centres. Assessment of current practices was required to inform appropriate recommendations for how an electronic tracking system can be incorporated into current processes. This section of the feasibility report will include the following

a. Overview of cargo movement - determining the corridors or routes, the type of movement (transhipment, bonded, internal) and the mode of movement (road or rail) as the focus of the study. Additionally, determining the key stakeholders in this movement -

such as warehouses, exporters, shipping lines, regulatory authorities, customs house agents, port authorities, transporters, and their individual and collective roles.

b. Overview of the regulatory conditions – customs law and practices that currently govern this movement. This includes determining the key bottlenecks hampering the efficiency of the private sector (practices such as customs escorting bonded goods, physical examination etc.) as well as determining which procedures are carried out online and offline (such as issuance of bonds, duty payments etc.)
c. Overview of the economic conditions - determining the economic significance of the movement. This would include determining the size of the economic sectors being impacted.

This section of the report has been compiled using data collected from different local stakeholders who are engaged in this movement in some capacity, including Customs officials from the Government of Sri Lanka as well as the private sector that is engaged in this movement. Information was collected from personal interviews, questionnaires and prior studies and reports.

II. Conceptualization of a tracking system for containerized cargo in the context of Sri Lanka

This section of the report will consist of the following:

a. How a tracking system would be incorporated into the current process, and which regulatory concessions/ incentives could be given to users as a result of this tracking system. Part I of the study will help us make better and more effective use of technology. The objective of this is two-fold:
   - The use of technology and its cost is accepted without opposition if there is an equal decrease in time and cost associated with the continuing processes.
   - And then implementing the technology in such a way that the bottlenecks are removed but still provide an equivalent alternative to the regulatory agency for monitoring/ security/ assurance through a digital and technological reform.

b. Assessment of different technologies available such as RFID and GPS. Each technology has its own benefits and limitations as well as varying costs of implementation. Volumes of movements, distance to be covered, type of movement studied in Part I will taken into consideration to determine which technology would be most suitable from a cost , security and visibility perspective. For example, one-time device make less sense for quick and short movements but tend to be more cost effective for longer movements.

c. Integration requirements - benefit and methodology to be followed in the integration of the tracking system with the legacy system currently prevailing. An analysis will be undertaken to understand the level of integration required and possible in part I of the study between the tracking system and the existing customs system to define the scope.

d. Determining the benefits of tracking such as benefits of visibility to key stakeholders, as well as highlighting any differences between technology options in terms of the benefits they would provide.

e. After assessing the different options available, a tracking system would be proposed. This tracking system would be fully conceptualized in the context of the containerized movement that is the focus of the study.

This section of the report has been completed with data compiled from different publicly available sources. We examined different case studies of how the incorporation of a tracking system could result in streamlining regulatory procedures and what other benefits resulted from such a project. Additionally, we compiled together the cost of implementation for different technologies such as RFID and GPS. Our prior experience in implementation of tracking systems as well as a study of global best practices will lend to the development of the proposed tracking system concept for Sri Lanka.

III. Estimated investment requirement

This section will be a quantitative assessment of the cost of implementation of the proposed tracking system. At this stage, only a high-level costing was compiled to meet the purpose of estimating budget allocation. This section of the report will include the following:

a. Tracking infrastructure - this would include hardware and software requirements for tracking such as GPS/ RFID devices, scanners, tracking platform etc. Cost comparison for other infrastructure such as cloud vs on premise would also be included.

b. Integration expenditure - cost and time involved for those who will be integrating data from the tracking system into the existing EDI/ASYCUDA system.

c. Monitoring infrastructure - this would include hardware and software requirements for monitoring movement such as control centers, first & rapid response teams etc.

d. Project scaling - High-level assessment of how the addition of new routes, movement types etc. could impact the investment required. It would highlight if there were any economies of scale that could be taken advantage of or if there would be exponential increase in investment from the expansion of the facility.

e. Cost accrual to key stakeholders - this would list which costs would accrue to which stakeholders and how the total investment would be divided among different stakeholders. The party which is accruing the benefit should usually be the one paying the cost. If the technology reduces the margins of custom house agents for the benefit of the importers, it will be impractical to have the cost borne by the agents and will be easier for them to oppose the reform.

IV. Assessment of project benefits

This section consists of a full assessment of project benefits, including benefits that accrue to different stakeholders from changes in regulatory reform/ procedural changes that are enabled through the proposed tracking system. This section of the report would identify and assess different economic and regulatory benefits of a tracking system, including:

a. Increased visibility

b. Streamlined regulatory processes/ or any other regulatory benefits offered to users of this system
c. Time-savings
d. Security
e. Availability of new data for analytics and future decision-making

Furthermore, this section of the report will evaluate the extent to which these benefits would be available to users of the tracking system.

2. Assessment of regulatory and economic environment in Sri Lanka

2.1. The Port Ecosystem

Port of Colombo is Sri Lanka's principal sea port. In 2019, Colombo Port handled 7.23 million containers of which 82% were transhipments. Therefore there is heavy container movement inside the port from the vessels to terminals and between terminals.

Colombo ports complex presently operates at three container terminals - Jaya Container Terminal (JCT), South Asia Gateway Terminal (SAGT) and Colombo International Container Terminal (CICT).

There are three customs inspection yards presently where customs inspections are conducted - Rank Container Terminal (RCT), Gray Line I container terminal (GL1) and Gray Line II container terminal (GL2).

1. Gray Line 1 – usually handles high-risk cargo
2. RCT – usually handles perishable cargo with low risk
3. Gray Line 2 – low to medium risk

Gray Line 1 and RCT are located Orugodawatta and Gray Line 2 is located in Grand Pass. Altogether there are 358 inspection bays

Stakeholders at Colombo Port:

(1) JCT is operated by Sri Lanka Ports Authority Sri Lanka Ports Authority (SLPA) is responsible for navigation operations, cargo handling and value addition such as transhipment, storage, warehousing, multi-country consolidation etc. SLPA handles LCL cargo inside the port at eight facilities under a special licence from customs and also at Peliyagoda yard.
(2) South Asia Gateway Terminal (SAGT) and Colombo International Container Terminal (CICT) are public-private partnerships.
(3) Sri Lanka Customs Department (SLCD)

2.1.1. Existing Customs Procedure

Sri Lanka Customs Department (SLCD) gets involved in the process from the moment the Custom Declaration form (CusDec) is submitted. Once the details in the CusDec is entered to the ASYCUDA system, the system decides the type of inspection needed and accordingly the place of customs inspection, based on the risk management criteria and the capacity of the inspection yards.

Existing customs procedure for imports –

1. Once container lands, customer gets a Bill of Lading (BL) and Delivery Order (DO) from the shipping agent. The DO marks the transfer of custody from shipping agent to customer
2. The customs declaration is filed on ASYCUDA by importer/CHA

3. The system assesses risk profile based and selects the type of inspection needed and allocated a place for inspection. This becomes visible to the CHA.
4. CHA/ Importer arranges for a truck to take the container from the terminal to the designated inspection yard
5. Before the container goes out of the port, the containers are scanned at the customs scanning facility inside the port. At the gate, the container gets a pass to proceed to the inspection facility.
6. Container information are entered into the yard operating system when a container enters at the main gate of the inspection facility. Upon entry, the container waits in a free parking space.
7. Gate pass and documents are submitted to customs and an officer is appointed for inspection
8. Container is moved to a free bay as per instructions by customs
9. Inspection can take about 15-20 minutes (low-risk cargo) and may take up to 3-4 hours depending on the type of inspection
10. Post-inspection, a release order is issued
11. Yard officer issues a yard release on the basis on the release order. After necessary payments, gate pass is issued
12. Gate pass and release order are presented at the gate to yard security and customs preventive officer, and the container is permitted to leave the yard
13. In the event the container inspection fails, the container is sealed, demounted and stacked in the stacking yard. The container is sealed.
14. In case of perishable goods, additional food inspections may be required.
15. In case of removal of goods is required, disposal may happen under inspection/ or alternatively disposal by auction

Current Issues in Customs Processes -

1. Road congestion between the port and inspection yards
2. Concentration of inspection demand to less than a 10-hour period
3. Inspection areas are scattered over several sites and often not easily accessible due to traffic. The scattered locations cause sub-optimal use of resources and planning. The customs is not able to control the truck flow as consignees themselves plan the truck move to the inspection areas.
4. A few shippers and consignees have been appointed to the green line which allows the container to pass directly without standard checks. Unfortunately, only a small part of the full container loads get the green line label, resulting in many inspections still today.
5. The total capacity on inspection is limited to about 1,000 containers per day.
6. Customs likes to implement 100% scanning (today about 70%). This will further increase the customs clearance times for containers, resulting in severe congestion in the port.
7. Customs has own detention areas near the inspection sites. Also in the port there are warehouses with goods under detention. The issue is here that these goods are not moved out frequently. Sometimes as auction are planned. The storage space in the port occupied by Customs is free of charge for Customs. This space is however very valuable for the port and should be cleared if possible

Customers Inspection Facility	RCT	Gray Line 01	Gray Line 02
Location Access	Colombo – Avissawella	Colombo – Avissawella	Sirimavo Bandaranaike M
	Low Level Road	Low Level Road	
Coordinates	6⁰56′37.86″ N,	6⁰56′36.80″ N,	6⁰57′22.80″ N
	79⁰53′03.28″ E	79⁰53′05.68″ E	79°52′31.10″E
Approximate Elevation (m amsl)	9	7	9
Land Extent (ac)	10	4.5	4.5
Distance from port (km)	7	7	5.6
Approximate travel time at 4 pm (min)	25	25	13
Approximate travel time at 6 pm (min)	20	20	7
Type of inspections	Perishable cargo, low-risk	Non-Perishable cargo	Low-medium risk

Customers Inspection Facility	RCT	Gray Line 01	Gray Line 02
Operating time	Round the clock	8.00 a.m. – 12 midnight	Round the clock
Peak time	6p.m. – 10p.m.	8:00 a.m. – 12 midnight	11a.m. -10 p.m.
Peak days	Wednesday, Thursday & Friday	Wednesday, Thursday & Friday	Wednesday, Thursday & Friday
Inspection Rate usual (TEU/day)	700-800	100 – 120	80-150
Existing capacity (TEU/day)	900	180	350
Number of Inspection bays	150	98	110
Staff Office	36	20	More than 10
Technical Staff		10	26
Supervisors/Steward	33	16	20
Labourers (temp)	100	120	150
Adequacy of Space	Limited number of inspection bays	Adequate	Adequate
Processing Time	Normal inspection 15-20 min	Single Consignment 30 mins	3-4 hours
	Detailed inspection 3-4 hours	Detailed inspection 2 hours	
Present Issues	Effect of the road congestion on container movement	Effect of the road congestion on container movement	No detail

Source: Inception Report

Overview of the regulatory conditions

The Sri Lanka Customs administration has been committed to introducing modern trade practices as part of the World Trade Organization (WTO) Trade facilitation agreement and with the support of other governmental and private sector stakeholders would be able to implement many best border practices that could improve the time release study performance metrics.

3. Conceptualization of a cargo tracking system in the context of Sri Lanka

3.1. Introduction to Electronic Cargo Tracking System

Electronic cargo tracking system, abbreviated as ECTS, is a technological tool for developing a secure, smooth, and efficient system for the movement of goods. The ECTS system consists of the following functionalities:

(1) GPS technology for live tracking of cargo,
(2) Electronic sealing technology so that any forcible unlocking or tampering can be detected,
(3) GPRS technology for communicating all trip events on a real-time basis, and
(4) Software application with graphic user interface for monitoring all cargo movement activity in real-time

These functionalities may be offered using a single device or multiple different devices together like GPS trackers, GPS locks, RFID seals etc. Different ECTS programs in different countries have utilized a variety of different electronic devices, depending on their specific use case factors such as existing infrastructure, planned investment, type of vehicles and movements covered under the program. Based on device selection, different ECTS programs may offer different benefits to their implementing agencies. But in general, most provide government agencies with the following key features:

(1) Real-time tracking of cargo and vehicles – allows for monitoring agencies to know the location of cargo and tracking at any given time. In case of any violations of the security protocols put in place, alerts can be generated in real-time
(2) Prevention of tampering or pilferage – any unauthorized unsealing or tampering can be detected
(3) Prevention of diversion – Similarly any diversion of the cargo or vehicles from their designated routes or unplanned long stops in transit also become visible
(4) Data and analytics – since cargo events are recorded from origin to destination, supply chain trends are identified and can be leveraged to make better, data-driven decisions.

Given these key functionalities of an ECTS system, ECTS programs add the most value in cases where there are non-physical barriers impeding international trade such as – (1) long regulatory and/or documentation processes, (2) high-security or sensitive cargo requiring security control and assurance, (3) movement of cargo is susceptible to diversion and/or

tampering (4) there is a requirement to address logistical or regulatory bottlenecks to reduce cost of logistics for businesses, or (5) there is lack of data or visibility of cargo flow. If an administration faces any of these issues, an ECTS system has the potential to bring substantial benefit in cargo security and visibility and creating a facilitative environment for international trade.

The UNESCAP study[1] on the application of technology for secure transport of goods under customs control also recommends the use of technologies such as satellite positioning systems, cellular communication systems, radio frequency identification (RFID) technology, geographical information systems (GIS) and web based software with enhanced visual graphics, to deal with major challenges in transport security. There are several notable examples of use of these technologies in the region. For example, the Customs authorities in China, Hong Kong and Thailand are using tracking systems based on these technologies to facilitate the movement of bonded goods between the Customs zones and for transport of containers from inland places to border crossings. India and Nepal have developed their own unique model of Electronic Cargo Tracking facilitated by SASEC program of the Asian Development Bank for the movement of transhipment cargo.

3.2. Assessment of Different Technologies

The two main technologies that are deployed to view and secure the movement of goods, containers or vehicles are GPS and RFID technologies. Both GPS and RFID devices have their own benefits and there are trade-offs in selecting one over the other. Each technology has its own benefits and limitations as well as varying costs of implementation. A suitable technology selection can be made on the basis of factors such as extent of visibility required, risk level associated with the monitored cargo (such as risk of tampering or diversion), site specific conditions such as volume of movement to be covered, distance to be covered.

Since most administrations usually consider the use of these two technologies to secure the movement of cargo, we have assessed both RFID and GPS based devices. Further by assessing Sri Lanka's specific use case and stakeholder input, we shortlisted 4 devices – RFID seal, GPS seal, Fixed GPS Tracker, Portable GPS Tracker. Sri Lanka's ECTS program can be designed to include either 1 or a combination of devices from this list.

The following table lays out the key differentiating features between 4 different kinds of electronic devices. The GPS seal and the GPS trackers, are based on GNSS/ GSM technology and the one-time RFID seal is based on RFID technology.

[1] https://www.unescap.org/sites/default/files/SCBM-fulltext_0.pdf

Parameter of comparison	GPS seal	GPS Asset Tracker	GPS Fixed Tracker	RFID seal
Images				
Requirements for device placement/ Installation	Seal can be placed on all containers or closed body truck doors. No installation is required.	This device does not have a sealing mechanism and can be placed inside any container or vehicle. No installation is required.	This device requires installation and will require to be hardwired into a vehicle. Can be used with vehicles only. Specialists required for installation who can connect the fixed tracker with the	Seal can be placed on all containers or closed body truck doors. No installation is required.

			vehicles main power supply and/or other systems.	
Visibility	Full visibility of the movement, with real-time tracking of the device	Full visibility of the movement, with real-time tracking of the device	Full visibility of the movement, with real-time tracking of the device	Limited visibility of the movement, as RFID seals only allow for milestone tracking. Milestones are referred to RFID readers that would be found along the journey.
Security	Real-time movement alerts in case of route deviation or unplanned stoppages on route. Real-time security alerts in case of any tampering, unauthorized unsealing of the cargo etc.	Real-time movement alerts in case of route deviation or unplanned stoppages on route.	Real-time movement alerts in case of route deviation or unplanned stoppages on route.	No real-time alerts. Visual inspection is required when the container/ truck arrives at the destination to detect any sign of tampering with the seal.
Access Control	GPS seals have controlled access, and only authorized personnel can seal/unseal this device. The details of the officer carrying out the sealing/unsealing process can be recorded.	No access control feature.	No access control feature.	No access control feature
Reusability	The GPS seals have battery, which have an active life of 3 days to 12 days depending upon the model of the seal.	Portable GPS trackers have battery, which have an active life of 3 days to 7 days depending upon the model of	Fixed GPS trackers are connected to the vehicle power source. Some models have a backup battery,	These one-time-lock seals (OTL) without requiring any power source but can be used one time only. Once used,

	These are rechargeable and can be charged for reuse after depletion of battery.	the tracker. These are rechargeable and can be charged for reuse after depletion of the battery.	providing up to 5-15 minutes of backup power in case of separation from their vehicle power source.	these are to be cut and discarded.
Portability	Portable devices can be placed on any vehicle or container.	Portable devices can be placed on any vehicle or container.	Installation required, these devices are hardwired into the vehicle that they are placed in	Portable device, can be used to seal any container or vehicle.
Data storage	Server linked only.	Server linked only.	Server linked only.	There are two system of using these RFID seals – one is embedding the data into the RFID seal's memory and the other is linking data with the unique TID of the tag and having the information accessed from a server. The former option of embedding information into the RFID tag embedded in the seal can help retrieve data in places where no internet is available. However, the limited memory available in RFID tags (which can store anywhere between 15 – 500 characters of data) cannot accommodate uploading of digital documents, images and so on. The server-based system requires internet

				but storage is of course limitless.
Communication	This seal is equipped with a GSM module (2G usually works well for such small data packets), and can send data in real time – about lock status, tampering, location, speed etc. In case of any breach, the system will give real time alert and location where the breach has occurred.	This device is equipped with a GSM module and can send data in real-time about its location, speed etc.	This device is equipped with a GSM module and can send data in real-time about its location, speed etc.	RFID readers are required to scan and read seals. The Readers are deployed at the point of destination where the seal is proposed to be scanned. Any tampering of the seal will only be detected at the time of scanning at the destination– as such RFID seals come with "passive" technology and there is no real-time information.
Costing	This device is one of the most feature-rich GPS devices with live movement monitoring and security features. As such, it is the highest price of all GPS devices compared in this table.	These devices are also economical, but due to the portability they are slightly higher priced than fixed GPS trackers.	These devices are the most competitively priced GPS products in the market.	The RFID seals are quite inexpensive in comparison to other options, small in size and have a long expiry. The RFID readers do increase the capital outlay considerably which has to be factored and divided by the number of seals scanned. In low volume environment, this solution can become expensive.
Uniqueness	Serial number or Device ID. The same device ID would be mapped to multiple trips/ movements as the device is	Serial number or Device ID. The same device ID would be mapped to multiple trips/ movements as the device is	Serial number or Device ID. The same device ID would be mapped to multiple trips/ movements as the device is	They have a visible serial number and an invisible embedded TID number, which is globally unique. As such,

	reused. Will need to rely on software platform for generating unique trip IDs for each trip/ movement.	reused. Will need to rely on software platform for generating unique trip IDs for each trip/ movement.	reused. Will need to rely on software platform for generating unique trip IDs for each trip/ movement.	each trip would have be mapped to a unique TID, as RFID seals are one-time use only.
Infrastructure required	These GPS seals can be used repeatedly after regular charging for which a charging facility is to be created at a central facility. These are reusable devices and such require return logistics operations of bringing back devices for charging once they are unsealed at the destination. Device data is sent to a server and can be viewed on a web or mobile application. Accordingly, related IT infrastructure will be required.	These GPS trackers can be used repeatedly after regular charging for which a charging facility is to be created at a central facility. These are reusable devices and such require return logistics operations of bringing back devices for charging once they are unsealed at the destination. Device data is sent to a server and can be viewed on a web or mobile application. Accordingly, related IT infrastructure will be required.	These GPS trackers are installed in the vehicle already, and as such no charging facility or return logistics required. Device data is sent to a server and can be viewed on a web or mobile application. Accordingly, related IT infrastructure will be required.	RFID seals also require RFID readers to read and record their scan in/ scan outs. Either portable handheld readers or fixed readers are required. Once seal is read by a reader, the reader then sends the data to a server and can be viewed on a web, reader or mobile application. Accordingly, related IT infrastructure will be required.
Customizations	The devices can be customized with printing of serial numbers/ QR codes/ barcodes of choice. Most GPS seals come with a silicone cover which can be customized to a color of choice	The device can be customized with printing or engraving of serial numbers/ QR codes/ barcodes of choice on the body of the device	The device can be customized with printing or engraving of serial numbers/ QR codes/ barcodes of choice on the body of the device	The RFID seal can be customized with printing of serial numbers/ QR codes/ barcodes on the seal body

	for easy color-coding			RFID seals can also be manufactured in different colors for easy color-coding

3.3. Benefits

This section will outline some of the key benefits associated with ECTS systems and how these benefits compare between the different technology options.

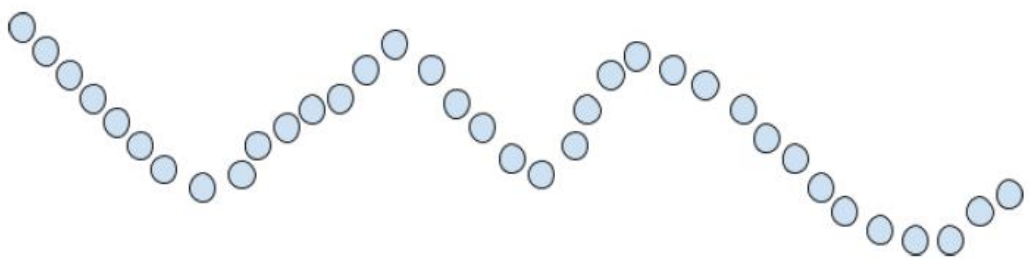
3.3.1. Visibility

The primary difference between RFID and GPS is visibility, and as such the benefit of going with any GPS device is going to be visibility.

RFID Location tracking :

We have NO INFORMATION about the actual movement of the vehicle between the two reader locations A and B. RFID tracking only gives you location and time stamp at reader location. Between A and B, it ASSUMES the route travelled. The vehicle might have deviated or stopped between the two readers but we will not know. RFID also does not give you any analytics. It assumes distance covered by mapping the distance between the readers without knowing the actual path the vehicle took between. It can also not tell you where the truck currently is and how long it will take to arrive at the next destination.

GPS location tracking :



With GPS tracking, you will have full information about the actual movement of the vehicle through the entire route. We know the exact route traveled by the vehicle and if any stoppage or diversion took place. With GPS tracking you can more accurately determine estimated arrival times (ETA) by assessing real-time conditions such as how long the truck was idle for and how long it was moving for. So you know when to expect the vehicle to arrive at the next destination.

On the other hand, with RFID you can only see when the vehicle arrived at the next destination. Post-movement you can see the time taken to cover the gap between the two readers, by comparing the two scan times. No information about what happened between them or efficiency.

3.3.2. Access Control

Among the 4 GPS/RFID devices that we have compared, only the GPS seal has the feature of access control. The GPS seal provides multiple sealing/ unsealing methods which allows there to be a complete audit trail of all operations that have been performed with the GPS seal. Every locking and unlocking event of the GPS seal are recorded with corresponding details such as – location and time, who performed the action, and which mode of operation was used.

(1) Multiple sealing/ unsealing methods for GPS seals

Mode of Operation	Description
Remote System (Web commands)	This method is built into many of the GPS seals in the market today. With a web command, the motor is activated to lock or unlock the device. The web command can be sent via a logged in user, leaving a full digital trace of who locked/ unlocked the device. It can be used to remotely operate the seal without the need of proximity – like from a central control room facility. Since there is a need to login to the application to be able to operate the lock, enhanced security features can be easily built into the application login/ user authentication process including biometric or OTP based authentication. Login can be mapped with each individual person. However, the use of this feature requires the GPS seal to be connected via internet to receive the command. As such, requires good data connection at the sealing/ unsealing point. Additionally, a central control/command room ops will be needed
Proximity Card (RFID)	Uses an RFID card (the better ones use 13.57Mhz i.e. High frequency instead of Low Frequency due to better security) The locking procedure brings about a sense of security to the operations/ on the ground personnel that are in charge of sealing/unsealing, as they can physically verify the lock status. Requires no use of any internet or network connection. Need physical proximity to open Restrictive as presence of authorized personnel is required. Loss of card can delay process of another card being mapped. Certain cards are “whitelisted” to be able to open a device. This whitelisting process can be tedious. It needs to be set via a command or during configuration. Loss of card can also pose a security risk if it not timely identified or reported. It is like loosing the keys.

Bluetooth	Double security - both proximity and application login access is required. GPS seal does not require to have a data connection, but the smart phone will require to have a data connection Each login will again be mapped with each person. For places where network is poor. Access to a smartphone with Bluetooth is required Proximity/ authorized personnel is required
SMS Protocol	ECTS device only needs to be in network and does not need to have data connection to perform the unlocking operation. Can set up a password to be sent via the SMS to authorise the opening. No digital record via an application as it goes directly via GSM - it will leave an imprint on the physical phone. Password setting for SMS commands is tedious and needs to be set via a command or during config.
Emergency	Can be cut open even when no battery in the device as no motor is locking it in place. No record of who opened. Need to have a bolt cutter to open the OTL

None of the other 3 devices offer any access control features. The device can be placed/ affixed by any authorized personnel without restriction or a digital trace.

3.3.3. Security

Both GPS and RFID based devices provide security features, but the extent of risk that is covered varies. As such it is important to consider the use case and associated risks to determine the best security device that is required.

<u>GPS Seal</u>

GPS seal offers the following security features -

(1) GPS seals provide end to end live tracking. Any route diversion, extended stoppages in transit can be detected in real-time.
(2) GPS seals have controlled access settings and as such every locking and unlocking event is detected. Any unauthorized locking or unlocking outside of the authorized geofenced area or

any tampering event (such as cutting of seal shackle or breaking of device) would also be detected in real-time.

(3) GPS seals can be configured to send push notifications for various parameters – such as overspeeding, tampering, locking/unlocking etc. Notifications can be sent via SMS, email or push notifications via app.
(4) Real-time alerts when entering or exiting a geofenced area

GPS seals would be best suited to the following conditions –

(1) When there is a centralized location for sealing and unsealing operations. Since a sealing and unsealing team will be required at these locations for the sealing and unsealing of device, charging etc.
(2) When there is a high risk of route diversion
(3) When there is a field team in place that can quickly intercept any real-time alerts for route deviation or tampering
(4) For particularly high-risk or high-value goods, or for goods with a history of pilferage, missing cargo etc.
(5) For closed body trucks or containers with doors
(6) When there is flexibility required in being able to secure and track any container or truck. Since it is a portable device, it can be affixed on any truck or container door.

GPS Portable Tracker

GPS portable tracker offers the following security features –

(1) GPS portable trackers provide end to end live tracking. Any route diversion, extended stoppages in transit can be detected in real-time.
(2) GPS portable tackers can be configured to send push notifications for parameters such as – overspeeding. Notifications can be sent via SMS, email or push notifications via app
(3) However, any device placement/ removal would not generate any real-time alerts and would have to be self-reported or evident after inspection of tracking data
(4) Real-time alerts when entering or exiting a geofenced area

GPS portable trackers would be best suited to the following conditions –

(1) When there is a centralized location for placement/ removal of devices. A centralized location will be required for charging and storing of the devices
(2) When there is a combination of different types of vehicles, containers or other cargo that require to be tracked
(3) When there is flexibility required in being able to track any container or truck. Since it is a portable device, it can be placed on any vehicle or container.
(4) When the risk of pilferable is low
(5) When there is high risk of route diversion
(6) When there is a field team in place that can quickly intercept any real-time alerts for route deviation

GPS Fixed Tracker

GPS fixed tracker offers the following security features –

(1) GPS fixed trackers provide end to end live tracking. Any route diversion, extended stoppages in transit can be detected in real-time.

(2) GPS fixed tackers can be configured to send push notifications for parameters such as – overspeeding. Notifications can be sent via SMS, email or push notifications via app
(3) One-time installation of the device is required. Thereafter, any removal of the device will generate real-time alerts
(4) Some models of fixed trackers also provide ability to remotely turn ignition off
(5) Real-time alerts when entering or exiting a geofenced area
(6) However, the tracking would be only of the vehicles and not of the containers. As such any detachment of the containers from the vehicles will not be detected.

GPS fixed trackers would be best suited to the following conditions –

(1) When there is a dedicated fleet of vehicles where the fixed trackers can be installed.
(2) When there is a combination of different vehicle types (closed body, open body)
(3) When the risk of pilferable is low
(4) When there is high risk of route diversion
(5) When there is a field team in place that can quickly intercept any real-time alerts for route deviation or detachment of fixed tracker from the vehicle

<u>RFID Seal</u>

RFID seals offer the following security features -

(1) One-time use seal, providing a unique identification number (TID) with each seal device. No two RFID seals can ever have the same the identification number as per global RFID standards. These TID numbers cannot be overwritten or replaced. As such, no RFID seal can be replaced without detection.
(2) Any tampering attempt would be visible upon manual inspection. However, it is important to note that when seals travel long distances on trucks or containers, there may be some external damage to the seal that may mimic an attempt of tampering.
(3) RFID seals are only readable by special RFID readers. Attempted tampering of the seal may leave these RFID seals unreadable.

RFID seals would be best suited to the following conditions –

(1) When there is no centralized location for sealing operations. This is because RFID seals can be easily procured or distributed across multiple locations. No specialized installation or equipment required. Any person would be able to easily lock a RFID seal on a container or truck door. RFID readers would only be required at the destination to verify the sanctity of the seal when it arrives at the destination.
(2) When there is no threat of route diversion. This is because RFID seals do not live-tracking, and as such any route diversion or extended stoppages in transit would not be detected in real time.
(3) When there is existing RFID reader infrastructure that can be leveraged. RFID reader infrastructure contributes to a majority of the infrastructure investment and is a fixed investment that would benefit from economies of scale.

Real-time movement alerts in case of route deviation or unplanned stoppages on route. Real-time security alerts in case of any tampering, unauthorized unsealing of the cargo etc.

3.3.4. Data Analytics

3.3.5. Ease of Use and Operations

3.4. Integration Requirements

Data received from any of the selected ECTS devices will require to be integrated with existing software platforms such as Asycuda. Integration of data for a consolidated view on a single platform supports quicker reviews and processes. Since different technologies offer different data, different integrations and process may be envisioned for different devices. Some examples of how data from ECTS devices can be integrated into existing software platform are as follows –

1. Gate passes can be issued automatically as vehicles enter the geofenced areas of the examination yards or are scanned in at the gate of the examination yard. This would be possible with all 4 types of devices.
2. For GPS seals for example, a report can be automatically generated that the GPS seal has not been tampered with and as such do away with the manual inspection of the seal at gate in, expediting the gate in processes. This would be possible with only the GPS seal.

3.5. System Design

This study combines experience from the existing use of electronic cargo tracking systems globally with requirements of the Sri Lanka administration. It aims to standardize the main features of the key technological components (both hardware and software) that are required for an effective electronic cargo tracking system to create a conceptual design for the system.

3.5.1. Hardware Features

(1) GPS seal – the primary purpose of this device will be to provide live tracking, real-time alerts and record every sealing and unsealing event. Specific features that should be provided in the selected GPS seal include –

- GNSS module for real-time tracking
- GSM/ GPRS module for real-time alerts
- Dual SIM preferred to maximize network coverage
- Multiple methods available for sealing/ unsealing
- Ability to configure data to transmit every 30 seconds to 15-minute intervals
- Wire or shackle
- Re-chargeable
- Battery type and size
- ISO certifications

(2) RFID seal – the primary purpose of this device will be to provide a unique identity to the trip being monitored as well as an additional layer of security
    - One-time use
    - TID based validation
    - Serial number visible on seal
    - Cable wire
    - ISO certifications

(3) RFID readers
    - Portable RFID reader
    - TID scanning to be enabled
    - GSM/ GPRS module to communicate scan details in real-time
    - GPS module to capture location of the scan

(4) Control room set up for monitoring
    - 8-panel video wall set up

3.5.2. Software Features

An overview of the software features that are recommended:

(1) Web-application
    - Data entry module to enter information about the trip being monitored. This can include data fields for goods information, importer/exporter on record etc.
    - Upload of supporting documents, images etc.
    - Payment gateway (if required)
    - Map UI to view current and historic tracking data. Geocoding required.
    - Control panel to remotely control
    - Configure geofences for authorized sealing, unsealing and stoppages (if any)
    - Dashboard view
    - Inventory management
    - Downloadable reports

(2) Reader application
    - UI to view details of scan
    - Mapping of RFID seal with

3.5.3. System Functionalities

This section of the report illustrates how the hardware and software will come together for the ECTS system to function:

(1) Device mapping
(2) Device Affixing
(3) Trip tracking
(4) Real-time alerts

4. Estimated Investment Required

4.1. Tracking Infrastructure

The tracking infrastructure includes hardware and software requirements for tracking such as GPS and RFID devices, scanners, tracking platform etc.

4.1.1. Hardware Infrastructure

Hardware requirements for different technologies will be different

<u>GPS seals</u>

Component	Price per Unit	Quantity Required	Total Amount
GPS seal + charger + accessories	USD 250	3,000	USD 750,000
Charging stations	USD 75	300	USD 22,500
Storage Crates and other equipment	USD 5	3,000	USD 15,000
SIM cards (dual sim for optimal performance)	USD 180	2,500	USD 540,000
Mobile phone for sealing/ unsealing operations/ trip mapping	USD 500	8	USD 2,000
Desktop for gate operations	USD 1,000	4	USD 4,000
Spares and replacements	USD 50	3,000	USD 150,000
Freight and duty (estimated)	@30%		USD 281,250
Total amount			USD 1,764,750
Per trip			USD 1.96 per trip

Notes:

(1) All cost analysis has been done over a 3-year period
(2) A total of 900,000 container trips over 3 years has been assumed
(3) Freight and duty combined is assumed to be 30%, applicable on the hardware cost and may vary at time of procurement

GPS Portable Trackers

Component	Price per Unit	Quantity Required	Total Amount
GPS portable tracker + charger	USD 90	3,000	USD 270,000
Charging stations	USD 75	300	USD 22,500
Storage	USD 5	3,000	USD 15,000
SIM cards (dual sim for optimal performance)	USD 90	3,000	USD 270,000
Desktop for gate operations	USD 1,000	4	USD 4,000
Spares and replacements	USD 20	3,000	USD 75,000
Freight and duty (estimated)	@30%		USD 114,750
Total amount			USD 7,71,250
Per trip			USD 0.86 per trip

Notes:

(1) All cost analysis has been done over a 3-year period
(2) A total of 900,000 container trips over 3 years has been assumed
(3) Freight and duty combined is assumed to be 30%, applicable on the hardware cost and may vary at time of procurement

GPS Fixed Trackers

Component	Price per Unit	Quantity Required	Total Amount
GPS fixed tracker	USD 65	3,500	USD 227,500
SIM cards	USD 90	3,500	USD 315,000
Desktop for Gate operations	USD 1,000	4	USD 4,000
Spares and replacements	USD 12	3,500	USD 42,000
Toolkit for installation	USD 25	3,500	USD 87,500
Freight and duty (estimated)	@30%		USD 82,050
Total amount			USD 758,050
Per trip			USD 0.84 per trip

Notes:

(1) All cost analysis has been done over a 3-year period
(2) Assuming 3,100 unique trucks are used for this movement, plus 400 trackers will be spare available for new trucks/ damages etc.
(3) Freight and duty combined is assumed to be 30%, applicable on the hardware cost and may vary at time of procurement

RFID Seal

Assumptions used

Component	Price per Unit	Quantity Required	Total Amount
RFID Seal	USD 3	900,000	USD 2,700,000
Desktop reader	USD 200	2	USD 400
Fixed readers[1]	USD 1,200	2	USD 2,400
Antennas[2]	USD 300	16	USD 4,800
Wiring + mounting equipment + connectors[3]	USD 600	2	USD 1,200
LED light indicators for scanning	USD 750	2	USD 1,500
LAN connections for the fixed readers + installation of network	USD 500	2	USD 1,000
Servers for readers	USD 1000	2	USD 2,000
Power outlets + power wiring and installation	USD 100	8	USD 800
Handheld readers for backup	USD 700	4	USD 2,800
Desktop for gate operations	$4,000	4	USD 4,000
Freight and Duty	@30%		USD 541,120
Total			USD 3,262,020
Total per container trip			USD 3.62 per trip

Notes:

(1) It is estimated that at minimum 2 readers will be required for Gate in and Gate out. In the case that additional readers will be required to capture scans through the journey of the container from the port to the examination yard, then additional readers would be required.
(2) Antenna quantity of 8 per reader is possible, allowing scanning to happen at multiple lanes
(3) This is just an estimation of the requirement and varies site to site
(4) All cost analysis has been done over a 3-year period
(5) A total of 900,000 container trips over 3 years has been assumed
(6) Freight and duty is assumed to a combined 30% of hardware costs and may vary at time of procurement

4.1.2. Software Infrastructure

All GPS Devices

Component	Total Amount
Standard web application	USD 324,000
Standard mobile application (for sealing/ unsealing and device mapping)	USD 108,000
Server Expenses (secure cloud server)	USD 75,000
Ongoing support and maintenance (L1 – L4)	USD 129,600
Total Amount	USD 636,600
Total Per Trip	USD 0.71 per trip

Notes:

(1) Depending on the extent of customization required, the software fee may vary
(2) Cost analysis is done on a 3 year period

RFID Seals

Component	Total Amount
Standard web application	USD 225,000
Standard mobile application	USD 25,000
Application for Handheld reader for seal scanning and data review	USD 50,000
Desktop Application for Fixed Reader for seal scanning and data review	USD 50,000
Desktop Application for Desktop Reader for seal scanning and data review	USD 25,000
Server	USD 75,000
Ongoing support and maintenance (L1 – L4)	USD 105,000
Total Amount	USD 555,000
Total Per Trip	USD 0.62 per trip

Notes:

(1) Depending on the extent of customization required, the software fee may vary
(2) Cost analysis is done on a 3 year period

4.2. Integration Expenditure

Any vendor may charge anywhere between USD 25,000 to USD 40,000 for integration with existing customs systems such as Asycuda or the port community system. This is just an estimate and may vary upon review of actual integration requirements.

4.3. Monitoring Infrastructure

<u>All GPS Devices</u>

Component	Quantity	Total Amount
Web application for monitoring		Already included above
Mobile application for monitoring		Already included above
Monitoring manpower (24x7 monitoring)	21 persons + 3 managers	USD 432,600
Central Control Room Equipment (videowall, desktops, phones)	Hardware equipment for a single facility	USD 55,000
Control Room management and overheads		USD 133,700
Interception team (optional)	12 persons + 3 managers	USD 286,200
Equipment for interception team including vehicle (optional)		USD 59,000
Interception team overheads (optional)		USD 86,300
Total Amount		USD 1,040,250
Total Per Trip		USD 1.16

Notes:

(1) Depending on staff availability locally, the amount may vary
(2) Cost analysis is done on a 3 year period
(3) Interception team is optional, but a recommended feature of ECTS system globally

<u>RFID Seals</u>

Component	Quantity	Total Amount
Web application for monitoring		Already included above
Mobile application for monitoring		Already included above
Monitoring manpower (24x7 monitoring)	12 persons + 3 managers	USD 286,200
Central Control Room Equipment (videowall, desktops, phones)	Hardware equipment for a single facility	USD 50,500

Control Room management and overheads		USD 133,700
Total Amount		USD 420,875
Total Per Trip		USD 0.47

Notes:

(1) Depending on staff availability locally, the amount may vary
(2) Cost analysis is done on a 3 year period
(3) Interception team is optional, but a recommended feature of ECTS system globally

4.4. Other Operational Infrastructure

GPS Seals

Component	Quantity	Total Amount
Sealing and Unsealing Staff	21 persons + 3 managers	USD 432,000
Asset Management Staff	12 persons +1 manager	USD 225,000
Overheads for staff management		USD 183,600
Hardware equipment for sealing and unsealing and asset management team		USD 51,000
Return Logistics	USD 0.40 per device trip	USD 360,000
Other operational overheads (such as fuel, electricity, internet etc.)	USD 0.50 per device trip	USD 450,000
Alerts (SMS and email alerts)	USD 0.10 per device trip	USD 90,000
Facilities for staff, device storage, charging etc.	USD 40 per Sq. Ft X 3000	USD 120,000
Total Amount		USD 1,892,250
Total per trip		USD 2.10 per trip

Notes:

(1) Depending on staff availability locally, the amount may vary
(2) Cost analysis is done on a 3 year period and for 900,000 container trips annually

GPS Portable Trackers

Component	Quantity	Total Amount
Sealing and Unsealing Staff	24 persons + 3 managers	USD 480,600

Asset Management Staff	12 persons +1 manager	USD 225,000
Overheads for staff management		USD 183,600
Hardware equipment for sealing and unsealing and asset management team		USD 51,000
Return Logistics	USD 0.40 per device trip	USD 360,000
Other operational overheads (such as fuel, electricity, internet etc.)	USD 0.50 per device trip	USD 450,000
Alerts (SMS and email alerts)	USD 0.10 per device trip	USD 90,000
Facilities for staff, device storage, charging etc.	USD 40 per Sq. Ft X 3000	USD 120,000
Total Amount		USD 1,953,000
Total per trip		USD 2.17 per trip

Notes:

(1) Depending on staff availability locally, the amount may vary
(2) Cost analysis is done on a 3 year period and for 900,000 container trips annually

<u>GPS Fixed Trackers</u>

Component	Quantity	Total Amount
Installation and Servicing Team	12 persons +1 manager	USD 225,000
Asset management Team	6 persons + 1 manager	USD 127,800
Overheads for staff management		USD 88,200
Hardware equipment for sealing and unsealing and asset management team		USD 27,000
Other operational overheads (such as fuel, electricity, internet etc.)	USD 0.50 per device trip	USD 450,000
Alerts (SMS and email alerts)	USD 0.10 per device trip	USD 90,000
Facilities for staff, device storage etc.	USD 40 per Sq. Ft X 1500	USD 60,000
Total Amount		USD 1,068,000
Total per trip		USD 1.19 per trip

Notes:

(1) Depending on staff availability locally, the amount may vary
(2) Cost analysis is done on a 3 year period

RFID Seals

Component	Quantity	Total Amount
Sealing/ Unsealing Team	21 persons +3 manager	USD 432,000
Asset management Team	6 persons + 1 manager	USD 127,800
Overheads for staff management		USD 88,200
Hardware equipment for sealing and unsealing and asset management team		USD 27,000
Other operational overheads (such as fuel, electricity, internet etc.)	USD 0.50 per device trip	USD 450,000
Alerts (SMS and email alerts)	USD 0.10 per device trip	USD 90,000
Facilities for staff, device storage.	USD 40 per Sq. Ft X 3000	USD 120,000
Total Amount		USD 1,386,750
Total per trip		USD 1.54 per trip

Notes:

(1) Depending on staff availability locally, the amount may vary
(2) Cost analysis is done on a 3 year period and for 900,000 container trips annually

4.5. Impact of Project Scaling

4.6. High-level costing

This high-level costing presented in the table below is purely a cost-based analysis. Median prices of industry standards for most components have been considered. It however does not include any margins, management or legal fees that would be additional in case of procurement from private vendors or if ECTS was offered as a service from a managed service provider. In such cases, it a 7-10% markup on these cost prices can be considered to be a suitable budget for the ECTS program in Sri Lanka for this movement.

	GPS Seal	GPS Portable Tracker	GPS Fixed Tracker	RFID seal
Hardware Expenses	1.96	0.86	0.84	3.62
Software Expenses	0.71	0.71	0.71	0.62
Integration Expenses	0.04	0.04	0.04	0.04
Monitoring Expenses	1.16	1.16	1.16	0.47
Other Operational Expenses	2.10	2.10	1.19	1.54
Total	USD 5.97	USD 4.87	USD 3.94	USD 6.29
	L3	**L2**	**L1**	**L4**

For Sri Lanka's use case, the recommendation would be to go with the GPS Fixed Trackers as lowest-cost option. Alternatively, the GPS seal may also be considered as offering the greatest level of benefits in terms of security and visibility benefits. The RFID seal has been eliminated as a feasible option for being the most expensive option with least visibility offered among the 4 device options available.

4.7. Cost Accrual to Key Stakeholders

5. Assessment of Project Benefits

The benefits from the application of ECTS technology may vary from case to case. In general, ECTS may significantly contribute to:

(1) Increase in safety and security en route;
(2) Reduction in transport time;
(3) Reduction in transport cost;
(4) Increased predictability about the location of the consignments;
(5) Increased efficiency of fleet management
(6) Tracking of container(s)/ vehicles in real-time from origin to destination
(7) Increase in safety and security control
(8) Reduced violations during the trip, such as smuggling incidents
(9) More visibility of movements of transport and goods
(10) Reduction in need for escort

The application of the ECTS program has tremendous potential to enhance the security of the transport process against robbery, pilferage, smuggling and will benefit both the control authorities and transport operators. Real time information on the status of vehicles and their integrity can help authorities take timely action and pre-empt such activities. Moreover, the fact that the trip is being monitored by the control authorities has a psychological impact on drivers and therefore may reduce the potential for collusion of drivers with people with nefarious designs to commit various crimes.

In addition, application of the electronic tracking system enhances safety. The real-time information about the status of the vehicle implies that in an event of an accident, the control authorities will know immediately and therefore take action to arrange timely emergency care to the victims. For traders and transport operators, this model can provide increased visibility of supply chain and increase predictability of consignments, which is particularly important for transport operators to manage their fleet and plan transport operations.

One of the main features of the ECTS system is its flexibility with regard to guarantee system. With real-time monitoring by the customs authorities, such guarantee may be exempted for

the users. In many cases, physical inspection. With the application of the model, such measures can be removed or reduced.

The use of electronic visibility and security systems as provided in this model allows carrier/shipper and Customs authority to know at all times where the goods are, and they will be immediately alerted if an intrusion/ exception event occurs.

The wide availability of these technologies coupled with relatively low costs associated with their application makes it possible to apply the model easily in different countries.